



Facebook to track popularity

Facebook is putting in systems in place to monitor accounts that get popular suspiciously quickly. <https://digit.in/jun20-102>



Samsung Money

Samsung has unveiled a debit card linked program that works with the Samsung Pay app. <https://digit.in/jun20-103>

FEATURE

probably not as waterproof or water-resistant as you think.

THE IP RATING SYSTEM DECODED

If a smartphone manufacturer markets a phone as being water-resistant or the obnoxious 'waterproof', chances are the smartphone sports an IP rating. If you ardently follow smartphone releases, you will know that this rating has something to do with protection against dust and water. However, there's much more to this than meets the eye. 'IP' stands for Ingress Protection or International Protection. It is essentially a massive body of standards that covers not just smartphones, but also other electronics, from electrical sockets (approximately IP22) to street lights (approximately IP44). The rating codes have no hyphens or spaces and simply consist of the letters 'IP' followed by, typically, two digits. The ratings are determined by the International Electrotechnical Commission, and according to this organisation, the rating system is "for classifying the degrees of protection provided by the enclosures of electrical equipment". In simple words, the rating determines how well a device can hold up against dust, water and other external pollutants.

The first digit following the 'IP' indicates the level of protection that the device's enclosure provides against the entry of foreign solid objects, which could range from anything between fingers to dust particles. There are six different levels of protection in this range. So, level 1 offers protection only against large objects while level 6 completely seals out dust from the device. Typically, gadgets such as smartphones and tablets have IP ratings above 5 or 6 in this category. This is necessary since smartphones need to have a level of protection against solids in order to safeguard the components and to keep camera lenses free of dust. The first table above indicates the level corresponding to the type of protection it offers.

The second digit represents the degree of protection against the entry of water into the enclosure of the

Level	Effective against	Description of protection
0	—	No protection against contact and ingress of objects
1	>50 mm 2.0 in	Protection against any large surface of the body, such as the back of a hand, but no protection against deliberate contact with a body part
2	>12.5 mm 0.49 in	Fingers or similar objects
3	>2.5 mm 0.098 in	Tools, thick wires, etc.
4	>1 mm 0.039 in	Most wires, slender screws, large ants etc.
5	Dust protected	Ingress of dust is not entirely prevented, but it must enter in sufficient quantity to interfere with the satisfactory operation of the equipment.
6	Dust-tight	No ingress of dust; complete protection against contact (dust-tight). A vacuum must be applied. Test duration of up to 8 hours based on airflow.

Level	Protection against
0	Not protected
1	Dripping water for 10 minutes
2	Dripping water when tilted up to 15 degrees for 10 minutes
3	Spraying water for 5 minutes at any angle up to 60 degrees from the vertical
4	Splashing water for 5 minutes
5	Water jets from a 6.3-millimeter nozzle for at least 3 minutes
6	Powerful water jets for at least 3 minutes
7	Immersion up to 1 meter for 30 minutes
8	Continuous immersion up to a depth specified by the manufacturer
9	High-pressure, close-range jets from any direction

device with levels ranging from 0 to 8. So, while levels 1 and 2 protect against varying degrees of dripping water, level 8 can handle full-fledged immersion beyond 1 meter and level 9 can handle extremely high-pressure, close-range jet sprays. The second table above details the water-resistance level and the corresponding type of protection.

Using the information in these tables you can gauge the water and dust resistance any device sports. Today, we usually see smartphones sporting an IP67 or IP68 rating. Let's break these down. In the case of the former, the device is completely dust-tight and can be immersed into water for up to one meter for a total duration of 30 minutes. IP68, on the other hand, means

that the device can be submerged even beyond 1 meter, up to the depth specified by the manufacturer. Also, while the time frame reads as 'continuous' here, some manufacturers still mention a time limit even if their device is rated IP68. Additionally, if you see an 'X' in either part of the rating, for example, IPX5, it means the rating for, in this case dust protection, isn't available. This doesn't necessarily mean it does not protect against that element (dust or water) but that the manufacturer has simply not tested that area. Sounds pretty simple right? It should mean that if you get a smartphone with an IP67 or IP68 rating you are protected against water and dust exceptionally well. Well, yes, but not quite...